Auditoría, Calidad y Fiabilidad Informáticas

(IT Auditing, Quality, and Reliability)

Universidad Complutense de Madrid Academic Year 2026/27

Practical 2: Introduction to specification languages II

Download the file `shop.maude`, understand it, and make the following changes:

Exercise 1 Define a module for lists of `String`. In this part it is enough to define the type and its constructors.

Exercise 2 Import the list module into the module `BST` and define a function `inorder` that, given a term of type `BST`, returns the list of `String` corresponding to the in-order traversal of the keys. Add any functions on lists you need to the module of the previous part.

Exercise 3 Use the `red` command to test that the function `inorder` works.

Exercise 4 Modify the specification so that buyers can purchase products one at a time. This way, even if the shop does not have all the products a customer needs, it can sell them as many as it has in stock.

Exercise 5 Make the same change so that the shop can acquire products from suppliers one at a time.

Exercise 6 Define new constructors for `Buyer` and `Supplier` (you do not need to delete the old ones) so that they indicate whether they want to deal in books or in films. Also define the new rules these constructors need.

Exercise 7 Define an example shop and use the commands `rew`, `frew`, `search`, and `show path` to test how the module works.